

32 channel NPN based DO

The IO modules communicate via RS485. The port can drive distances up to max 200 meters without the use of any repeater (*this feature however also depends on the signal strength of the Modbus Master Device*).

The RS485 Digital IO module is sturdy, has low power usage and is easy to use.

32 Port DO Module: -



The IO modules are mounted on DIN rail mountable casing and with exposed connectors and LED indicators. Baud rate and slave ID can be changed by software setting.

The design of the modules incorporates '**resettable Fuses**' to safeguard against reverse polarity connection both for **Power** and **Communication** port.

Specifications

General –

I/O Connectors	5.08 mm pitch pluggable screw Terminal Block
Dimensions	157 mm L x 110 mm B x 73 mm H
Power	Input Power –24 V DC options Typical – 12V DC @ 150mA
Operating Temperature	0 – 60° C (32 ~ 140°F)
Storage Temperature	-20 - 70° C (-4 ~ 158°F)
Storage Humidity	5 ~ 95 % RH, non – Condensing
Output mode	Collector NPN output
Response speed	0.01 seconds (10ms)
Application design	Industrial-grade anti-interference design, stable

Certifications



Additional Features: -

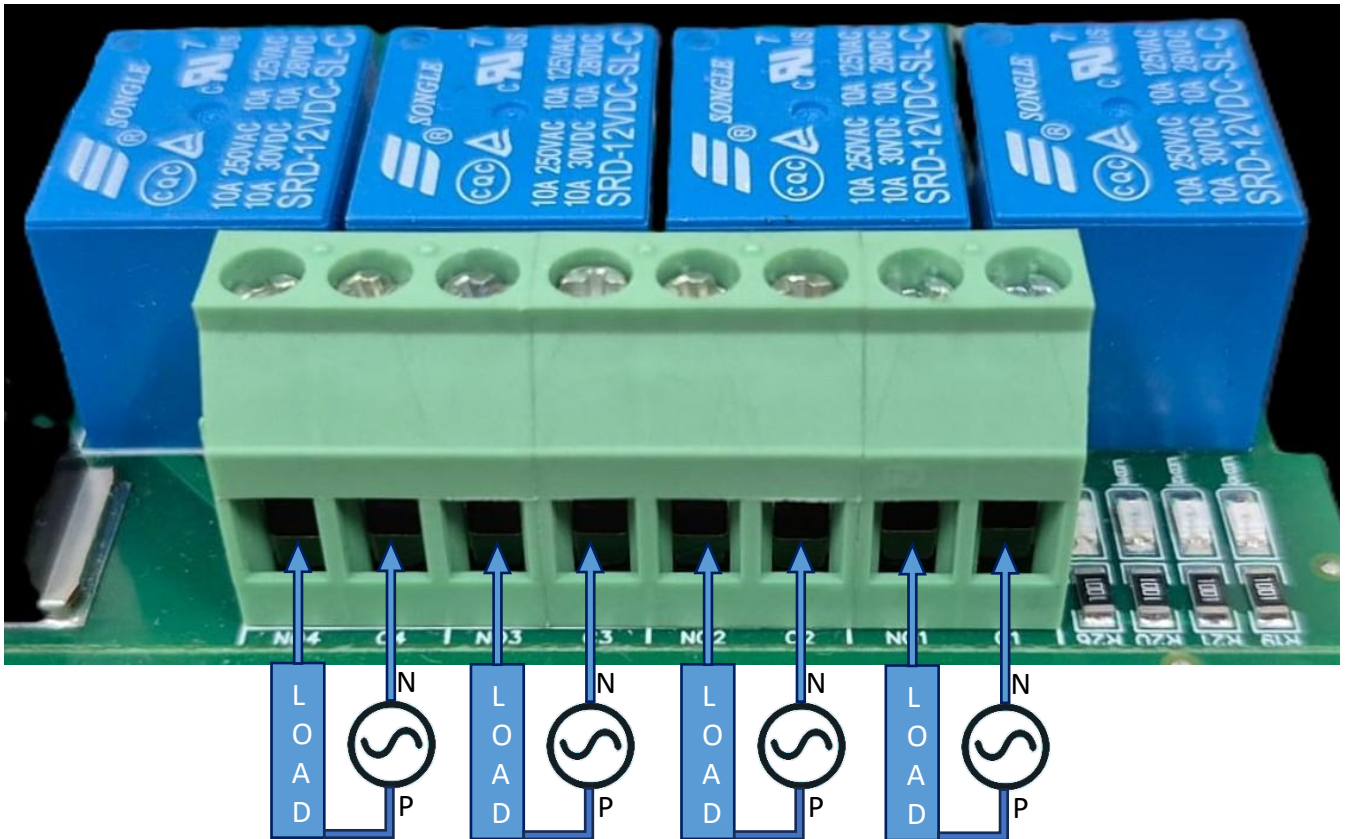
Communication ports are isolated
Input power reverse polarity safety
ESD Safety IEC 61000-4-2, ± 30KV contact, ± 30KV air
EFT IEC 61000-4-4, 50A (5/50ms)
400V isolation.
CRC Error check.
No configuration needed on the IO board

Configuration Settings: -

Communication Speed	9600 – 115200 Kbps (Software selectable)
Data Bits	8
Parity	None
Stop bit	1
CRC	Yes
Slave ID	Software selectable
Function code	0x01
DO Register Address	0 to 31

ID	Function Description	Register Description	Protocol	Data Type
1	DO 1	1	RS485	1 Bit Boolean
2	DO 2	2	RS485	1 Bit Boolean
3	DO 3	3	RS485	1 Bit Boolean
4	DO 4	4	RS485	1 Bit Boolean
5	DO 5	5	RS485	1 Bit Boolean
6	DO 6	6	RS485	1 Bit Boolean
7	DO 7	7	RS485	1 Bit Boolean
8	DO 8	8	RS485	1 Bit Boolean
9	DO 9	9	RS485	1 Bit Boolean
10	DO 10	10	RS485	1 Bit Boolean
11	DO 11	11	RS485	1 Bit Boolean
12	DO 12	12	RS485	1 Bit Boolean
13	DO 13	13	RS485	1 Bit Boolean
14	DO 14	14	RS485	1 Bit Boolean
15	DO 15	15	RS485	1 Bit Boolean
16	DO 16	16	RS485	1 Bit Boolean
17	DO 17	17	RS485	1 Bit Boolean
18	DO 18	18	RS485	1 Bit Boolean
19	DO 19	19	RS485	1 Bit Boolean
20	DO 20	20	RS485	1 Bit Boolean
21	DO 21	21	RS485	1 Bit Boolean
22	DO 22	22	RS485	1 Bit Boolean
23	DO 23	23	RS485	1 Bit Boolean
24	DO 24	24	RS485	1 Bit Boolean
25	DO 25	25	RS485	1 Bit Boolean
26	DO 26	26	RS485	1 Bit Boolean
27	DO 27	27	RS485	1 Bit Boolean
28	DO 28	28	RS485	1 Bit Boolean
29	DO 29	29	RS485	1 Bit Boolean
30	DO 30	30	RS485	1 Bit Boolean
31	DO 31	31	RS485	1 Bit Boolean
32	DO 32	32	RS485	1 Bit Boolean

DO



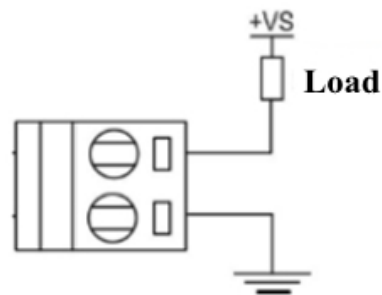
- **Digital Output (DO) operation:** Each channel provides two terminals: **C (Common)** and **NO (Normally Open)**. When the relay coil is energized, terminal **C** is internally connected to **NO**, allowing current to flow. When the relay is de-energized, the connection between **C** and **NO** remains open, and the load stays OFF.
- **Load Wiring Connections:**
 - **AC Load Connection:** The **AC Live (Phase)** wire should be connected to terminal **C**. The load is connected between terminal **NO** and **Neutral (N)**.
 - **DC Load Connection:** The **positive supply (+V)** should be connected to terminal **C**. The load is connected between terminal **NO** and **Ground (GND)**.
- **Crucial Safety Recommendation and Rating:** It is strongly recommended to switch the Live (Phase) conductor in AC applications to ensure the load is safely de-energized when the relay is off.

Note: Ensure that the connected load does not exceed the relay's contact rating of **10A @ 250VAC** (or **10A @ 125VAC**) or **10A @ 30VDC** (or **10A @ 28VDC**).

Recommended Cable Electrical Characteristics: -

22 AWG Cable	Shielded Twisted pair should be used
Tinned Copper	Recommended
Nominal Conductor DCR	14.7 ohm / 1000 ft
Nominal Capacitance	11 pf / feet (conductor to conductor)
High Frequency Non-Insertion Loss	0.5db / 100f

NPN Wiring



Application design as above

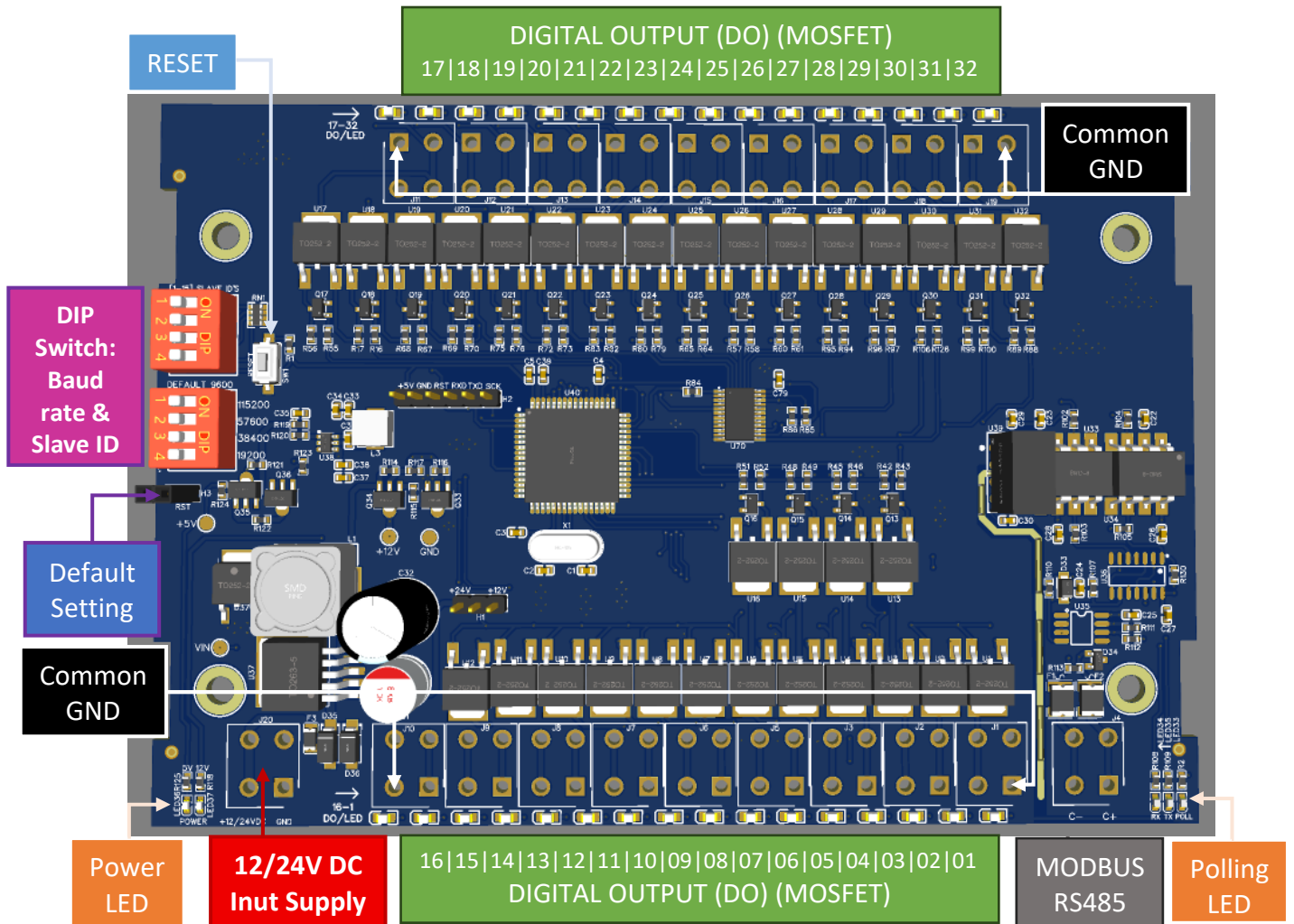
03: SR = 1000ms

Name	00010
Default Baudr...	960
Selectable Baud	0
Default SlaveID	1
Selectable Slave	0
Default Parity	3
Selectable Pari...	0
Default Stopbit	1
Selectable stop	0

This is default Modbus setting that is Baud rate is 9600, Slave ID is 1.

Baud rate is multiple of 10($960 \times 10 = 9600$)

To change baud rate and slave ID , at the selectable register the required value of Baud rate and Slave ID, Make a power cycle (Power ON/OFF the product) and Connect Modbus with changed parameters.



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